Zijia Li

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Education

University of Pennsylvania — Philadelphia, PA
M.S.E. in Data Science GPA: 3.95/4.00 — Aug. 2025 – May 2027 (expected)

University of Nottingham — Nottingham, UK
B.S. in Mathematics with Applied Mathematics — Sept. 2023 – June 2025
Joint program with **University of Nottingham Ningbo China** — Sept. 2021 – June 2023
GPA: 88/100 (UK system), 3.97/4.00 (WES)

Research Interests

I am interested in **world models**, **visual representation learning**, video models, and image generation, with an emphasis on learning generalizable visual structure and dynamics for prediction, planning, and control.

Research Experience

University of Pennsylvania — Philadelphia, PA
Graduate Researcher, GMLR Lab, GRASP Lab — Supervisor: Prof. Jiatao Gu

NFWM: Normalizing Flows are Capable World Models — May 2026 – Present
- Developed NFWM, an end-to-end **normalizing-flow world model** that jointly learns an invertible visual encoder and action-conditioned probabilistic dynamics through exact maximum likelihood.
- Enabled latent-space planning, **policy learning in imagination**, and observation generation within one model through information-preserving representations and distributions over possible futures.
- Validated planning and policy learning across deterministic and stochastic environments, with gains over JEPA-based baselines in stochastic planning and over behavior-cloned policies in Atari Boxing.

Generative Latent Dynamics with Frozen Visual Representations — Dec. 2025 – May 2026
- Explored JEPA-style world models using frozen DINOv2 features, a VAE, and **flow matching** for both next-state and joint action–state prediction.
- Implemented Monte Carlo rollouts, Q-function-based planning amortization, and CEM and gradient-based planning to study uncertainty and the computational cost of iterative sampling.

Equilibrium Matching (EqM) for Generative Modeling — Oct. 2025 – Dec. 2025
- Reproduced EqM ImageNet baselines and explored empirical connections with flow matching.

Zhejiang University — Hangzhou, China
Research Assistant, Intelligent Computing and System Lab — Supervisor: Prof. Qinming He

Flow Matching and Diffusion Sampling — Aug.–Oct. 2024 and May–Sept. 2025
- Tested MeanFlow variants based on different definitions of average velocity. Ran small-scale parallel-sampling experiments and studied rectified flow, DDPM, DDIM, and DPM-Solver.

University of Nottingham — Nottingham, UK
Undergraduate Researcher, Mathematical Sciences — Supervisors: K. van der Zee and L. Espath

Physics-Informed Neural Networks — Feb. 2024 – Oct. 2024
- Implemented **physics-informed neural networks** in JAX for wave equations in spherical coordinates and ordinary differential equations, with investigation of boundary-condition difficulties.

Publication

Normalizing Flows are Capable World Models. [Project page]
Zijia Li*, Yong-Hyun Park*, Shusheng Yang, Jiatao Gu. *Equal contribution.
Workshop version accepted at CoRL 2026 LEAP Workshop. Full manuscript under review.

Awards and Honors

Dean's Scholarship (top 5%), UNNC	2023
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Zhejiang Provincial Scholarship	2022–2023
Honorable Mention, Mathematical Contest in Modeling (MCM)	2023
Third Prize, Formula Student Electric China	2022

Skills

Methods: World models, visual representation learning, normalizing flows, flow matching, diffusion models, Transformers, VAEs, model-based reinforcement learning, model predictive control.
Programming and tools: Python, R, MATLAB, SQL, JAX, LaTeX.
Languages: Chinese (native), English (proficient).
TOEFL iBT: 109 (Speaking: 26) GRE: 326 (Analytical Writing: 3.5).